

Revision implementation and readiness report

Swarm and Evolutionary Computation | 4 October 2026

Manuscript: Baseline Configuration and Random-Sampling Controls in Metaheuristic Comparisons for Wind Farm Layout Optimization

Decision: substantially improved, but further author work is required before submission. The revised documents compile and the supplied revision records can be audited. The archive still cannot substantiate complete research reproduction or independent strict feasibility of all legacy layouts.

Audit disposition	Count	Meaning
Fully addressed	5	The stated correction is implemented and verified at its specified scope.
Partially addressed	7	Supported edits or checks are implemented; missing evidence, new experiments or author facts remain.
Supplied optimization records	11,640	All available revision CSV families pass stored-record checks.
Missing original optimization records	36,190	Required to reproduce primary and earlier supporting analyses.

What the updated ZIP contains

Revised modular and generated single-file manuscripts, rebuilt main and supplementary PDFs, revised cover letter and declarations, five highlights, unchanged CSVs, new audit and focused-check scripts, derived reliability tables, machine-readable verification, an author checklist and a complete change log. The original ZIP-readiness report is retained separately as a pre-edit assessment.

Evidence boundary

No original optimizer was rerun, and no missing objective, coordinate, p-value, module, DOI or author confirmation was invented. Existing scientific statistics are retained as historical analyses from the supplied text and tables. The newly computed Lillgrund endpoint is explicitly distinct from the original signed-rank endpoint.

Disposition of all twelve audit areas

A1. Conflicting manuscript versions — Fully addressed

Incorporated: Recovered the fuller later scientific source into every modular section; generated the single-file manuscript from that canonical source; rebuilt both PDFs and refreshed supplementary references.

Remaining: Future edits must follow the canonical-source workflow.

A2. Incomplete reproduction package — Partially addressed

Incorporated: Added a working standalone archive audit, focused checks, build script, environment pins, coverage inventory and honest reproduction instructions. Replaced unsupported “all checks pass here” wording with historical attribution.

Remaining: Supply 36,190 original optimization records, diagnostic records, core implementation, original analyses and environment.

A3. Coordinate precision / strict feasibility — Partially addressed

Incorporated: Quantified rounded-coordinate failures and Lillgrund AEP replay differences; retained original labels/objectives. Five future writers now preserve 17 significant digits, with binary round-trip checks.

Remaining: Recover original unrounded final positions or rerun affected experiments. Strict archived feasibility is not independently repaired.

A4. PSO variance proposition — Fully addressed

Incorporated: Main and supplementary statements now say zero limit variance iff $c_1 \cdot c_2 \cdot (p - g) = 0$; $p = g$ is equivalent only when both coefficients are positive. Added proof explanation and checked the zero-coefficient counterexample and both PSO settings numerically.

Remaining: No experimental rerun needed for this algebraic correction.

A5. Conditional ranking versus reliability — Partially addressed

Incorporated: Explicitly named qualification-then-conditional quality; added Lillgrund success counts/Wilson intervals and paired all-run superiority with bootstrap intervals, exact sign tests and a 16-comparison Holm family. Explained the 17/30 versus 30/30 reversal.

Remaining: Apply an agreed all-run endpoint to all original benchmark/site cases after supplying the missing records; recover precise feasibility evidence.

A6. Inference estimands and calibration — Partially addressed

Incorporated: Removed new-farm generalization from group sensitivity, described case exchangeability as hypothetical, clarified existing case weights and cross-case seed dependence, and limited the intersection-union claim to calibrated constituent tests. Bootstrap equivalence is described as approximate interval inclusion.

Remaining: Original inference code/data are needed for joint seed resampling, null calibration and any preregistered or population-level claims.

A7. Visibility of expanded baselines — Fully addressed

Incorporated: Added GA to the main Horns Rev table, GA and both analytic-gradient variants to main IEA37 best-run results, an IEA37 means/feasibility table, and the nine-method benchmark summary. Historical eight-method ranks remain explicitly separate.

Remaining: Stronger DE/CMA-ES experiments remain an extension, not a result claimed here.

A8. Analytic-gradient interpretation — Fully addressed

Incorporated: Explained analytic objective AND constraint derivatives, different hybrid restart choices, piecewise smoothness, uncharged constraint/solver costs and limitations of batch timings. Added the `c_g=19` accounting bounds as saved-prefix arithmetic, not a new run.

Remaining: An objective-gradient-only ablation would be new research; the revised text does not claim it was done.

A9. Evaluator transfer versus re-optimization — Fully addressed

Incorporated: Identified finer-rose/Gaussian/power-curve studies as re-evaluation of saved layouts; identified spacing and gradient studies as direct re-optimization; retained the reversal at 6D.

Remaining: Optimization under alternative evaluators is an optional new experiment and is not implied by the current results.

A10. Constraint handling / baseline fairness — Partially addressed

Incorporated: Added implementation-specific scope, penalty-unit and repair differences, and limits of the single DE configuration.

Remaining: Run normalized-penalty/repair controls and stronger or tuned baselines before claiming algorithm-family superiority.

A11. Declarations and author facts — Partially addressed

Incorporated: Synchronized manuscript and separate declaration text with the supplied roles/funding/interest statements; disclosed ChatGPT assistance; replaced unsupported open-repository promises with the archive's actual status; updated cover letter/highlights.

Remaining: Authors must confirm all facts, affiliations, roles, AI wording, manuscript approval and submission eligibility. No public DOI has been invented.

A12. Presentation and submission package — Partially addressed

Incorporated: Rebuilt PDFs, refreshed references, added legible compact main tables, concise highlights, a shorter cover letter, a starting guide and this implementation record. Archived original loose notes separately.

Remaining: The main manuscript remains long and dense; authors should choose any further shortening. Journal acceptance and editorial formatting are not guaranteed.

New analysis: Lillgrund all-run reliability

The primary qualification-then-conditional-quality rule can prefer a method with fewer successful runs. At 6,030 evaluations, PSO-VNS has 17/30 feasible labels and mean AEP 113.074 GWh conditional on those successes; MS-SLSQP has 30/30 and 112.865 GWh. Reporting both success and conditional quality makes this distinction visible.

A feasible label wins over an infeasible label. Two feasible runs compare AEP with a 10^{-9} GWh tie tolerance. Two infeasible runs tie. The score $(\text{wins} + \text{ties}/2)/30$ is accompanied by a 95% percentile interval from 20,000 resamples of matched seed pairs, and an exact sign test over discordant pairs. Holm correction includes eight opponents at each of two budgets, for 16 contrasts.

Budget	PSO-VNS vs.	W / T / L	Score	95% interval	Holm p
6030	PSO	17 / 13 / 0	0.783	[0.700, 0.867]	0.000168
6030	MS-SLSQP	10 / 0 / 20	0.333	[0.167, 0.500]	0.395
30030	PSO	28 / 0 / 2	0.933	[0.833, 1.000]	1.22e-05
30030	MS-SLSQP	28 / 0 / 2	0.933	[0.833, 1.000]	1.22e-05

The 6,030-evaluation comparison against MS-SLSQP is not significant under this corrected sign test. The original reported signed-rank result is a different statistic and remains identified separately. These intervals concern seed variation on the fixed Lillgrund block, not variation across sites. All endpoints use original feasibility labels and do not repair rounding uncertainty.

Legacy precision: findings and prospective fix

The stated feasibility tolerance is 10^{-6} m, while archived coordinates preserve only 0.001 m or 0.01 m. Among labelled-feasible records, rounded rechecks fail for 236 exact-gradient, 78 Laplace, 229 spacing, 17 low-budget Lillgrund and 73 high-budget Lillgrund runs. Horns Rev boundary verification remains unavailable without its model.

The maximum AEP replay difference from rounded positions among labelled-feasible large-budget Lillgrund runs is 0.225072 GWh (PSO, seed 18). The revision retains original values and labels and explains why rounded-layout failure cannot establish that the unrounded predecessor was infeasible. Future writers preserve 17 significant digits; original positions must still be recovered or rerun.

Independent verification and coverage

CSV family	Records	Stored-record result
ga	2,040	PASS
gahr	60	PASS
gaiea	120	PASS
grad	240	PASS
laplace	2,160	PASS
spacing	6,480	PASS
lg16	270	PASS
lg16b	270	PASS

Checks verify complete numbered shard sets, unique run keys, finite scalar values, declared evaluation budgets, objective-wake-loss identities and gradient charging identities. They do not reproduce missing original experiments or their statistics. All 140 original CSV hashes match the upload.

Focused verification covers 2,006 exact binary float round-trips, rejection of invalid/nonfinite coordinates, exact sign-test reference values, a Holm reference family, Wilson boundary symmetry, the two PSO second-moment spectral radii and the degenerate-coefficient zero-variance counterexample. Every provided and new Python source passes syntax compilation; optimizer behavior is not rerun.

Document	Pages	Verification
SWEVO_cover_letter.pdf	1	Compiled; no visible TBD/??
SWEVO_declarations.pdf	1	Compiled; no visible TBD/??
SWEVO_manuscript.pdf	48	Compiled; no visible TBD/??
SWEVO_manuscript_full.pdf	48	Compiled; no visible TBD/??
SWEVO_supplement.pdf	58	Compiled; no visible TBD/??

The modular and generated single-file main PDFs have identical extracted text page by page. Final TeX logs have no undefined or multiply defined references, overfull boxes or LaTeX warnings. Page rendering and document layout inspection are part of the final package checks. Machine-readable evidence resides in validation/.

Absent project modules

authors_objective, authors_optimizers, extra_baselines, feasible_init, hornsrev_model, hybrid_lxssa_bvns, iea37_experiments, iea37_model, init_hook, mpce_experiments, mpce_inference_extra, mpce_results, original_vns, rs_vns, wflop_model.

This is a directly identified dependency list, not a guarantee that supplying only those files will restore every historical pipeline. Historical 200+ claim checks and external-model/diagnostic validations remain author-reported until their actual scripts and data are supplied.

Readiness and author actions

1. **Recover the complete research archive.** Add the 36,190 original optimization records, the diagnostic outputs, original model/optimizer modules and all original analysis/check scripts. Supply an environment specification that actually runs them. Rerun the historical checks and regenerate all original tables from that archive.
2. **Recover full-precision coordinates.** Add unrounded coordinates and their record keys for affected legacy runs, or undertake documented reruns. Verify constraints and objective replay at declared tolerances before calling the archive reproducible. Do not infer unrounded coordinates by projecting the saved rounded layouts.
3. **Complete benchmark reliability analysis.** Decide on an all-run endpoint, apply it to the full original data, and report success uncertainty and performance together. The new Lillgrund endpoint is executable, but original benchmark records are missing.
4. **Review inferential claims.** Decide whether to retain the approximate interval-based equivalence sensitivities. Stronger claims require calibration and appropriate dependence treatment; the revised manuscript explicitly limits population interpretation.
5. **Confirm author details.** Existing manuscript affiliations, corresponding-author emails and names were preserved. Confirm their current correctness. Do not substitute affiliations from unrelated prior projects without author approval.
6. **Confirm declarations.** CRediT roles, no-grant funding statement and no-conflict declaration came from the uploaded declarations file. Confirm them for every author. AI disclosure now names the ChatGPT assistance used in this revision; review the description and verify the final scientific content yourselves.
7. **Confirm submission eligibility.** Confirm author approval, originality, concurrent-submission status, third-party/data permissions and any required statements in the actual submission system. The revised cover letter does not invent approvals. Complete a public data/code deposit and insert its real identifier only when available.
8. **Choose any new experiments or shortening.** Normalized constraint-handling controls, stronger tuned DE/CMA-ES, null calibration, optimization under alternative evaluators and larger/full farms cannot be claimed as completed. The manuscript remains extensive; choose further shortening based on the journal's current instructions and editorial priorities.

Assessment for the target journal

The revised focus is experimental methodology for comparing swarm and evolutionary methods: baseline configuration, controls, budget accounting, reliability and the limits of equivalence claims. This is a plausible thematic fit. The main outstanding scientific risks are incomplete reproducibility evidence, strict-feasibility uncertainty, conditional-performance selection and dependence/calibration assumptions. Editorial acceptance cannot be predicted from formatting or manuscript length alone.

The current package is appropriate for author review and further completion. Submit only after the author-dependent evidence and declarations are resolved, or after the journal/editor explicitly accepts the disclosed archive limitations. No numerical acceptance probability is assigned.

Build and audit workflow

Read `START_HERE.md`. Run `experiments/audit_archive.py` and `experiments/check_revision.py` from the archive root. Build the documents with `latex_source/analysis/build_swevo.sh`. Edit modular section files and regenerate `SWEVO_manuscript_full.tex`; keep a single scientific version. The audit-only environment is pinned in `requirements-audit.txt`; the missing original optimizer environment is not reconstructed.